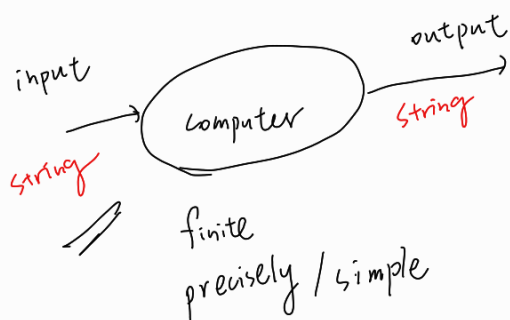


Ch6 Language vs. F.S.M

machines { finite state machine / sequential machine
combinational circuit / gating network



A	0	A	1	B	1	0
B	0	A	0	A	1	1
						1
0	A	1	B	1		

Language: This is a book.

Σ : nonempty finite set of symbols

$\Sigma = \{0, 1\}$

$\Sigma = \{a, b, c, d, e\}$

power of Σ

$$\Sigma^0 = \Sigma$$

$$\Sigma^k = \Sigma^{k-1} \cdot \Sigma^1$$

$$\Sigma^{n+1} = \{xy \mid x \in \Sigma, y \in \Sigma^n\}$$

Σ^k 組起來長度為 k 的字串

$$\Sigma^0 = \{\lambda\}$$

$$w = xy \quad w = x\lambda$$

$$ab = ab\lambda$$

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots \cup \Sigma^n$$

$$\Sigma^+$$

Σ

power of Σ

$$\begin{cases} \Sigma^1 = \Sigma \\ \Sigma^k = \Sigma^{k-1} \cdot \Sigma^1 \\ \Sigma^{n+1} = \{xy \mid x \in \Sigma, y \in \Sigma^n\} \end{cases}$$

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots \cup \Sigma^n$$

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \dots \cup \Sigma^n$$

$$\|w\| \rightarrow \text{長度} \quad \|\lambda\| = 0$$

$$\|xy\| = \|x\| + \|y\|$$

prefix
postfix (suffix)

substring

$\Sigma = \{0, 1, 2\}$ alphabet

$\Sigma^+ = \{0, 1, 2, 01, 00, 02, \underline{10}, \underline{11}, \underline{12}, 20, 21, 22, 000, 001, \dots\}$ word

closure

$$\|w\| = ? \quad \begin{cases} w = \lambda & \|w\| = 0 \\ w = xy & \|w\| = \|x\| + \|y\| \end{cases}$$

$$x = x_1 \cdot x_2 \cdot \dots \cdot x_n$$

$$y = y_1 \cdot \dots \cdot y_m$$

$$x \cdot y = xy = x_1 x_2 \dots x_n y_1 y_2 \dots y_m$$

$$\|xy\| = \|x\| + \|y\|$$

$$\Sigma = \{0, 1\}$$

$$x = 01$$

$$x^2 = 0101$$

$$x^0 = \lambda$$

$$x^1 = 01$$

prefix 前序字符串

$$w = x \cdot y$$

postfix 后序字符串

substring 子字符串

abcde

a
ab
ab
abc
abcd
abcde

f
of
def

$$w = xyz \Rightarrow x$$

proper substring 子字符串, 但跟原串不一样 (可能有 λ ??)

a given alphabet Σ

any subset of Σ^* is called "language"

$$\Sigma = \{0, 1\} \quad A = \Sigma \rightarrow$$

$$(finite) A = \{0, 01, 001\}$$

$$B = \{1, 01, 0001, 00001, \dots\}$$

把所有可能字符串

$$L = \Sigma^*$$

$$L \subseteq \Sigma^*$$

$$\Sigma = \{a, \dots\}$$

$$\Sigma^* = \text{"1st" } \dots$$

$$L \subseteq$$

$$B \subseteq \Sigma^* \quad \text{方法 } s+u+v \Rightarrow \text{regular expression}$$

$$A^0 = \{\lambda\}$$

$$A^1 = A$$

$$A^{n+1} = \{ab \mid a \in A^1, b \in A^n\}$$

$$A^+ = UA^n$$

$$A^* = \bigcup_{n=0}^{\infty} A^n = A^+ \cup \{\lambda\}$$

regular expression

describe string

even length

odd length

$$\Sigma = \{x, y\}$$

$$A = \{xx, xy, yx, yy\} = \Sigma^2$$

$$A^* =$$

$$\text{dir } a^* \cdot \log$$

任何字符串

$$\{x, y\}^*$$

$$\{x^3y^2xy\} \{x^2y^3\}$$

$$\{x^2y^3\} \{x^2y^3\}$$

$$\{x^2y^3\} \{x^2y^3\} \{x^2y^3\}$$

$$C = \bar{1} + \bar{2} \quad \textcircled{=}$$

$$= ()$$

这样描述

$$(x+y) \left((x+z) \times 3 + 4 \right) \leftarrow (5 \times 6)$$

$$((())) ()$$

$$* + A = \frac{()}{()}$$

